

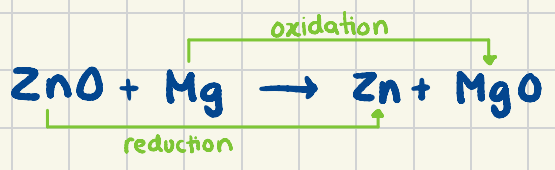
6 - Electrochemistry

15/12/24

Oxidation & Reduction

- oxidation is the gain of oxygen by an element
- reduction is the loss of oxygen by an element
- * oxidation & reduction always take place together

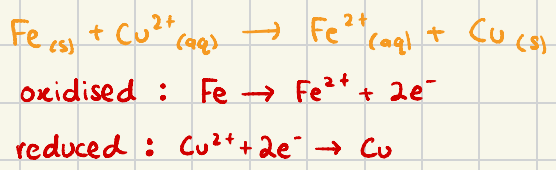
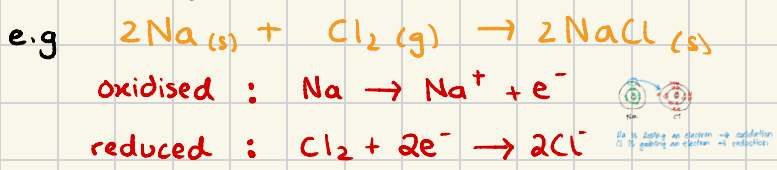
Oxidation	Loss of e^-	Reduction	Gain of e^-
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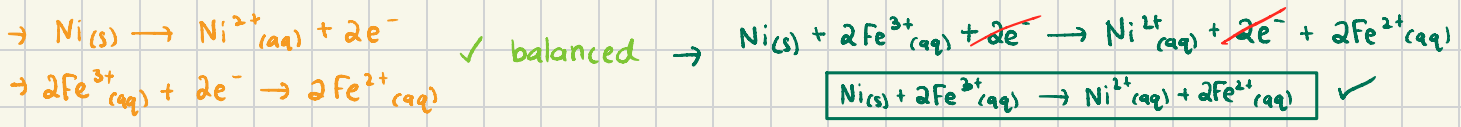
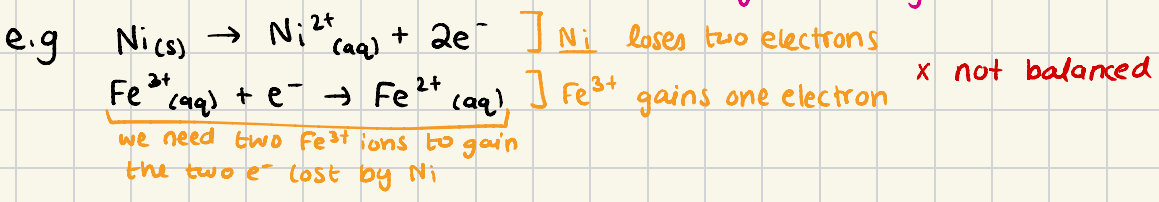
REDOX REACTION

Half-equations

→ Separating a reaction into two separate reactions, one showing oxidation & the other reduction



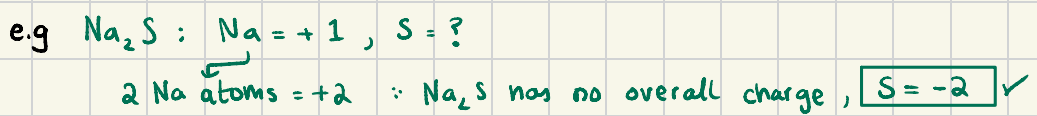
⇒ we can make balanced ionic equations by balancing the half-equations



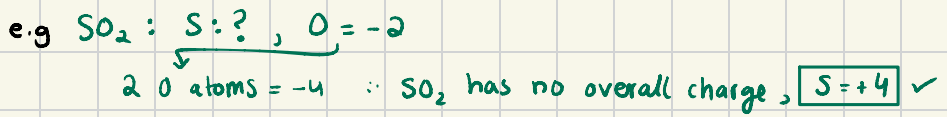
Oxidation number [abbreviation : ox. no.]

- a number given to each atom/ion in a compound that shows us its degree of oxidation
- can be +, -, or 0
- higher +ve number means more oxidation, higher -ve number means more reduction

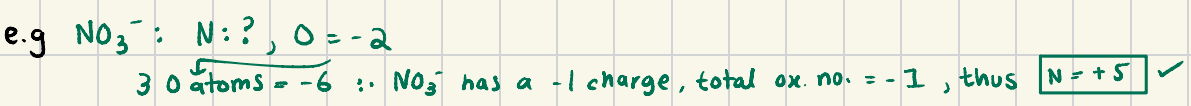
* a metal with a non-metal : the metal has the +ve ox. no. & the non-metal has the -ve ox. no.



* a non-metal with a non-metal : the more electronegative element is given the -ve ox. no.



* compound ions : use rule number 5 to work out the ox. no. we don't know



⇒ oxidation number rules

- 1) oxidation number of an uncombined element is zero
- 2) in compounds, many atoms/ions have fixed oxidation number:
 - ↳ Group 1 elements are always +1
 - ↳ Group 2 elements are always +2
 - ↳ Fluorine is always -1
 - ↳ hydrogen is always +1 (except in metal hydrides, where its -1)
 - ↳ oxygen is always -2 (except in peroxides, where its -1 & in F_2O , where its +2)
- 3) oxidation number of a monatomic ion is the same as its charge
- 4) sum of oxidation number in a compound is zero
- 5) sum of oxidation number in an ion is equal to its charge
- 6) the more electronegative element is given the negative oxidation number

⇒ oxidation numbers are written as roman numerals

e.g. $CrO \rightarrow$ chromium (II) oxide, $Cr_2O_3 \rightarrow$ chromium (III) oxide, $CrO_3 \rightarrow$ chromium (VI) oxide

Oxidising & reducing agents

→ oxidising agent: oxidises another substance but itself is reduced

- increases the oxidation number of another atom/ion
- when that happens, its own oxidation number decreases

e.g. oxygen, chlorine, potassium manganate (VII)

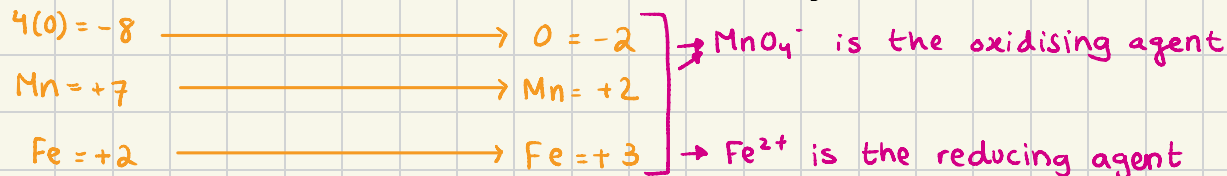
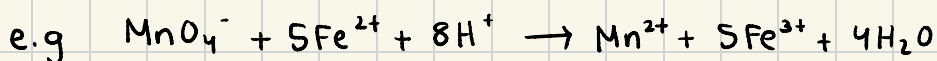
→ reducing agent: reduces another substance but itself is oxidised

- decreases the oxidation number of another atom/ion
- when that happens, its own oxidation number increases

e.g. hydrogen, potassium iodide & reactive metals (such as aluminium)

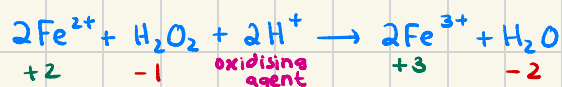
* oxidation is an increase in oxidation number

* reduction is a decrease in oxidation number

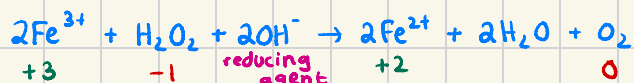


* many substances can act as either oxidising or reducing agents depending on the substances they react with as well as the conditions

e.g. hydrogen peroxide



OR

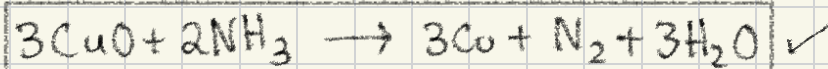
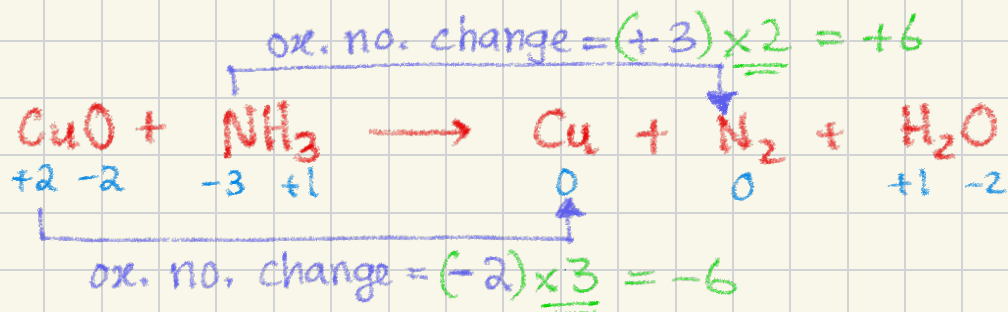


Constructing equations using oxidation numbers:

- 1) identify oxidation number changes
- 2) balance oxidation number changes
- 3) balance charges
- 4) balance atoms

Copper (II) oxide $[CuO]$ reacts with ammonia $[NH_3]$ to form copper, nitrogen $[N_2]$ and water.

Deduce the balanced equation:



Disproportionation reaction

* a reaction in which the same element is simultaneously oxidised & reduced

e.g

